Which Parts of Conformal Evaluation Survive on the Simplex? Validity, Efficiency and Conditional Coverage for Composition-Valued Predictions

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Predictions that are compositions, such as softmax vectors, topic proportions or vote distributions, can be given conformal regions on the simplex, but it is unclear how such methods should be evaluated and which scores suit which conditions. We study the evaluation of split conformal regions for composition-valued responses with seven scores (Euclidean, total variation, Hellinger, KL, Linf, Aitchison, Dirichlet likelihood), in plain, locally normalised and Mondrian variants, plus CV+. We verify that the finite-sample coverage law of split conformal prediction is score-free (a Beta law, checked for three scores and two noise families), so marginal coverage cannot discriminate between scores; we show that on class labels the total-variation, Hellinger and KL scores yield exactly the LAC label sets; and we calibrate binned conditional-coverage metrics against a null simulation in which conditional coverage is exact. In a synthetic grid of 48 configurations with a known conditional law (100 replicates each), the ranking of scores by region mass depends on the noise law (the Aitchison region is 2.5 times the Euclidean one for Dirichlet noise at $K = 3$ but 0.9 times for rounded logistic-normal noise at $K = 10$), local normalisation repairs conditional coverage only under heteroscedasticity and never down to the null floor, Mondrian calibration by argmax is unusable at 30 calibration points, and Aitchison region sizes change several-fold with the zero-replacement constant. A first-order Dirichlet theory predicts pointwise Euclidean conditional coverage (RMSE ≤ 0.010) but fails for the Aitchison score. Experiments on CIFAR-10 and CIFAR-10H and a covariate-shift check complete a reproducible protocol; we claim no new guarantee and limit conclusions to the tested designs.

1 Introduction

Many predictors output a *composition*: a non-negative vector that sums to one, such as a softmax probability vector, a vector of topic proportions or an annotator vote distribution. A user of such a prediction needs more than a point estimate; they need to know whether the attached uncertainty statement can be trusted. Split conformal prediction offers a finite-sample marginal coverage guarantee under exchangeability for any score function [Angelopoulos and Bates, 2021], and has been applied to label sets of classifiers [Angelopoulos et al., 2021, Ding et al., 2023], to batch group-conditional guarantees [Jung et al., 2022] and, for composition-valued responses, to regions of the simplex [Javanmardi et al., 2024]. What is missing is an agreed, reproducible way to *evaluate* such methods when the response lives on the simplex: the efficiency measure of a region of the simplex, the strata used to probe conditional coverage, the number of calibration points at which these diagnostics can be trusted, and which of the observed differences between scores are real.

We therefore treat evaluation itself as the object of study and ask: *for composition-valued predictions, which parts of conformal evaluation are score-free guarantees, and which depend on the geometry of the score, the calibration scheme and the sample size?* We study a response $y \in \Delta^{K-1}$ and a regression $x \mapsto \hat{p}(x) \in \Delta^{K-1}$, and compare split-calibrated regions built from seven scores (Euclidean, total variation, Hellinger, KL, Aitchison/CLR, Linf and a Dirichlet negative log-likelihood), in plain, locally normalised and Mondrian variants, plus CV+. A synthetic oracle with a known conditional

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law provides ground truth for conditional coverage; CIFAR-10 softmax outputs and CIFAR-10H vote proportions give a real label-set bridge and a real composition-valued response.

Contributions. (i) A concrete evaluation protocol for composition-valued conformal regions, with explicit choices of size measure, strata and uncertainty quantification (Section 3.2). (ii) An exact check, per score and family, of the finite-sample coverage law of split conformal prediction (Section 3.3, Section 4). (iii) An empirical comparison of scores and calibration schemes, with confidence intervals over repeated calibration sets and data seeds, and an oracle measurement of conditional coverage (Section 4). (iv) A null simulation that quantifies how large a difference in binned conditional-coverage metrics must be to be interpretable at realistic sample sizes. (v) An explicit statement of where the conclusions hold and where they must not be extrapolated (Section 5). We claim no new coverage guarantee and no conditional guarantee; our novelty claim is limited to the literature we read (Section 2).

2 Related work

Conformal prediction and its evaluation. Split (inductive) conformal prediction calibrates a score threshold on held-out data; its coverage conditional on the calibration set follows a Beta law, and Angelopoulos and Bates [2021] recommend comparing the empirical distribution of coverage over calibration draws with this law to separate implementation errors from fluctuation, and stress that adaptivity must be assessed separately from coverage. For classifiers, Angelopoulos et al. [2021] introduce regularised adaptive sets and size-stratified coverage violation (SSCV), and Ding et al. [2023] evaluate class-conditional coverage gaps (CovGap) and set size over repeated splits when classes have few calibration points. Jung et al. [2022] give group-conditional guarantees for any score and group collection; we do not run them (Section 5). These works treat the softmax output as a score input and do not use the geometry of the simplex.

Multivariate and constrained responses. Feldman et al. [2023] calibrate multi-output quantile regions and compare region volume; Principato et al. [2024] show for responses satisfying linear equality constraints that projecting the point prediction onto the constraint set cannot increase an A-norm score, so regions shrink without losing coverage. Sum-to-one is such a constraint, but the simplex has additional inequality constraints $y \geq 0$ that their equality-constraint result does not address.

Composition-valued responses. The closest setting is Javanmardi et al. [2024], who conformalise credal sets: the response is an annotator label distribution and the output is a split-conformal region of the simplex defined through total variation, Wasserstein, KL, an inner-product score or a Dirichlet likelihood. Efficiency is measured as the fraction of a grid of the simplex covered (resolution 0.005, $K = 3$ in their main experiment), which does not scale to larger K. They report average coverage, but do not compare calibration schemes, probe conditional coverage, or use log-ratio geometry. We reuse the setting and the scores as a starting point and add the evaluation questions above.

Compositional data analysis. Aitchison [1982] models compositions through log-ratios; the Aitchison distance is the Euclidean distance between centred log-ratio (CLR) coordinates and requires strictly positive parts. Zeros, which occur whenever labels are one-hot or components fall below a detection limit, need a pre-declared replacement rule; we make that rule an experimental factor.

3 Setting, methods and evaluation protocol

3.1 Setting and conformal regions

Let (X, Y) have $Y \in \Delta^{K-1} = \{y \in \mathbb{R}^K_{\geq 0} : \sum_k y_k = 1\}$, and let $\hat{p} : \mathcal{X} \to \Delta^{K-1}$ be a predictor trained on a fit split. Given a score $s(y, p)$ (larger = worse fit), a calibration set of size n from a split independent of the fit split, and level α, the split conformal threshold is $\hat{q}$, the k-th smallest calibration score with $k = \lceil (n+1)(1-\alpha) \rceil$ ($\hat{q} = \infty$ if $k > n$), and the region is $C(x) = \{y \in \Delta^{K-1} : s(y, \hat{p}(x)) \leq \hat{q}\}$.

Scores. With $\tilde{y} = \mathrm{closure}(y + \varepsilon)$ for zero replacement: Euclidean $\|y - p\|_2$; total variation $\frac{1}{2}\|y - p\|_1$; Hellinger $\|\sqrt{y} - \sqrt{p}\|_2$; $\mathrm{KL}(y\|p)$; $\|y - p\|_\infty$; Aitchison $\|\,\mathrm{clr}(\tilde{y}) - \mathrm{clr}(\tilde{p})\|_2$; and the Dirichlet negative log-density $-\log \mathrm{Dir}(\tilde{y}; \hat{\phi}\tilde{p})$ with a single precision $\hat{\phi}$ fitted by maximum likelihood on a separate split.

Calibration schemes. *Plain* split conformal; *normalised* scores $s/\hat{\sigma}(x)$ where $\hat{\sigma}$ is a random forest for the score fitted on a third split; *Mondrian* calibration with one threshold per predicted-dominant-part group; and *CV+* with 10 folds for the Euclidean score and a multinomial-logit predictor.

3.2 Evaluation protocol

Marginal validity. For each configuration we repeat the whole pipeline (fresh data draw, fit, calibrate, test) R times and report the mean test coverage with a confidence interval over replicates, and compare the empirical distribution of calibration-conditional coverage with the Beta law below.

Size. A region's size is its Lebesgue measure on the simplex relative to the measure of the simplex, estimated by uniform Monte Carlo draws on Δ^{K-1} for $K \leq 5$ only; for larger K the hit fraction vanishes and no volume is reported. Sizes under different reference measures (for example an Aitchison ball) are not compared with each other.

Conditional coverage. Coverage is estimated within strata of the predicted maximal part and of the predicted entropy (quantile bins), within a boundary stratum ($\min_k y_k < 0.01$), by worst-slab coverage [a slab chosen on one half of the test data and evaluated on the other; Romano et al., 2020], and, in the heteroscedastic worlds, by the true noise scale. On synthetic data these are measured against a known conditional law. We report the range of bin coverages (min/max), and calibrate their interpretability by a null simulation (Section 4).

3.3 Theory

Proposition 1 (Coverage law of split conformal prediction)**.** *Let the calibration scores $S_1, \ldots, S_n$ and the test score S_{n+1} be exchangeable with continuous common distribution function F, independent of the fit split, and let $\hat{q} = S_{(k)}$ with $k = \lceil (n+1)(1-\alpha) \rceil \leq n$. Then the coverage conditional on the calibration set, $V = \mathbb{P}(S_{n+1} \leq \hat{q} \mid S_{1:n}) = F(S_{(k)})$, has the law* $\mathrm{Beta}(k,\, n+1-k)$*. In particular $\mathbb{E}V = k/(n+1) \in [1-\alpha,\, 1-\alpha+\frac{1}{n+1})$ and the law does not depend on F, hence not on the score, the predictor or the response family.*

Proof. $F(S_i)$ are i.i.d. uniform by the probability integral transform, and the k-th order statistic of n i.i.d. uniforms is $\mathrm{Beta}(k, n+1-k)$. The mean follows from $k \geq (n+1)(1-\alpha)$ and $k < (n+1)(1-\alpha) + 1$. □

This is the finite-sample statement used to check the pipeline [Angelopoulos and Bates, 2021]. It requires continuity of the score distribution: with exact zeros or rounded parts, ties can occur and V stochastically dominates the Beta law (coverage $\geq 1-\alpha$ but possibly conservative). Group-wise (Mondrian) calibration obeys the proposition within each group, with n replaced by the random group size. Normalised scores stay valid because $\hat{\sigma}$ is fitted on a separate split. CV+ carries only a weaker guarantee ($\geq 1 - 2\alpha$ in general) and is not covered by the proposition.

Invariances. Sets are unchanged under any strictly increasing transformation of the score (for instance Euclidean distance and its square), and the Aitchison ball is the same in every orthonormal ILR basis, since the Euclidean distance of CLR coordinates equals that of any orthonormal coordinates in the $(K-1)$-dimensional hyperplane. Only the *shape* of the score matters for size and conditional coverage, not its monotone reparametrisation.

Proposition 2 (Vertex labels)**.** *If the response is a vertex e_j (a class label), then total variation, Hellinger and* $\mathrm{KL}(e_j\|p)$ *are strictly decreasing functions of p_j alone ($\frac{1}{2}\|e_j - p\|_1 = 1 - p_j$, $\|\sqrt{e_j} - \sqrt{p}\|_2^2 = 2 - 2\sqrt{p_j}$, $\mathrm{KL} = -\log p_j$), so the sets they induce coincide with those of the LAC score $1 - p_j$. The Euclidean (Brier) score $\|e_j - p\|_2^2 = 1 - 2p_j + \|p\|_2^2$ and the Aitchison score depend on the rest of p, and generally give different sets.*

This identifies which simplex scores are genuinely different from LAC on classifier outputs.

Pointwise conditional coverage under Dirichlet noise (large precision). If $Y \mid x \sim \mathrm{Dir}(\phi\,\mu(x))$ and the predictor is $\mu(x)$, then for large ϕ the residual is approximately Gaussian with covariance $\Sigma_{\mathrm{L2}} = (\mathrm{diag}\,\mu - \mu\mu^\top)/(\phi+1)$ in raw coordinates, $\Sigma_{\mathrm{Ait}} = G\,\mathrm{diag}(\mu)^{-1}G/(\phi+1)$ in CLR coordinates ($G = I - \mathbf{1}\mathbf{1}^\top/K$), and $\Sigma_{\mathrm{Hel}} = (I - \sqrt{\mu}\sqrt{\mu}^\top)/(4(\phi+1))$ in Hellinger coordinates. Writing $\lambda_j(x)$ for the eigenvalues of $\Sigma(x)$, the squared score is approximately $\sum_j \lambda_j(x) Z_j^2$ with Z_j i.i.d. standard normal; the marginal threshold q is the $(1-\alpha)$ quantile of the x-mixture of these laws, and the predicted conditional coverage at x is $\mathbb{P}(\sum_j \lambda_j(x) Z_j^2 \le q^2)$. We check this first-order approximation by simulation in Section 4 and report where it fails; it is an approximation, not a theorem.

4 Experiments

All numbers below are produced by the scripts listed in the reproduction notes (Appendix A) from the committed result files; $\alpha = 0.1$ is the primary level. A *replicate* is an independent draw of data, predictor fit, calibration split and test set; intervals are 95% t-intervals of the mean over replicates unless stated.

4.1 Setup

Synthetic worlds (E1). $x \sim \mathcal{N}(0, I_4)$, mean $\mu(x) = \mathrm{softmax}(Bx + b)$ with a fixed random (B, b) per configuration, $K \in \{3, 5, 10\}$. The conditional law is Dirichlet ($y \sim \mathrm{Dir}(\phi\mu)$, $\phi = 20/h(x)$), logistic-normal ($y = \mathrm{softmax}(\log\mu + 0.5\,h(x)\varepsilon)$), multinomial proportions ($30/h(x)$ counts, hence exact zeros) or a rounded logistic-normal with parts below 0.01 set to zero. The noise factor is $h \equiv 1$ (homoscedastic) or $h(x) = \exp(0.8\,w^\top x)$ (heteroscedastic). Predictors are a multinomial-logit regression on soft labels (the correct mean model for Dirichlet and multinomial noise, misspecified for the logistic-normal) and a random forest. Each replicate uses 1000 points to fit, 500 to fit $\hat\sigma$ and $\hat\phi$, n calibration points (default 500) and 2000 test points.

Real data. E2: a small CNN trained on CIFAR-10, with the test set split at random into calibration ($n = 1000$) and test points, 200 splits; label sets from the scores evaluated at the vertices. Two checkpoints (2 and 12 epochs; accuracy 0.740 and 0.881) vary the predictor's quality.

4.2 Validity is score-free: the Beta law (C1)

For each of three scores and two noise laws ($K = 5$, linear predictor) we draw 3000 calibration sets of size $n \in \{19, 50, 100, 500\}$ and evaluate the calibration-conditional coverage on a test pool of $2 \cdot 10^6$ points. Fig. 1 shows the empirical mean and 5/95% quantiles against the Beta band of Proposition 1. The largest deviation of an empirical mean from the Beta mean over the 24 cells is 0.002; Kolmogorov–Smirnov p-values are 0.03–0.98 (Dirichlet) and 0.02–0.86 (multinomial), with 3 of 24 below 0.05, consistent with chance given 24 tests of which none is corrected. Note that this check uses the continuous noise laws; with exact zeros (E3, rounded worlds) ties only give the lower bound.

4.3 Efficiency of the regions depends on the score and on the noise law (C2)

All 15 main variants and CV+ are marginally valid in all 48 configurations (mean coverage 0.896–0.909), so differences in size compare equally valid regions. As efficiency measure we use the region *mass* `pool_frac`: the share of the marginal law of y inside the region, estimated on the test pool. It is defined for every K, unlike the Lebesgue volume, which we compute only for $K \le 5$ (Appendix, Fig. 5; the Spearman correlation between mass and volume is 0.89 at $K = 3$ and 0.96 at $K = 5$). Fig. 2 gives the mass relative to the plain L_2 region of the same replicate, for heteroscedastic worlds with a linear predictor. Three patterns stand out. (i) The ranking of scores is not stable across noise laws. The plain Aitchison region is 2.45, 1.98 and 1.45 times the plain L_2 region for Dirichlet noise at $K = 3, 5, 10$, but 0.90 times for the rounded logistic-normal at $K = 10$. (ii) The Dirichlet NLL score gives the smallest region in every Dirichlet panel (0.82, 0.67, 0.62 of L_2); the information-based

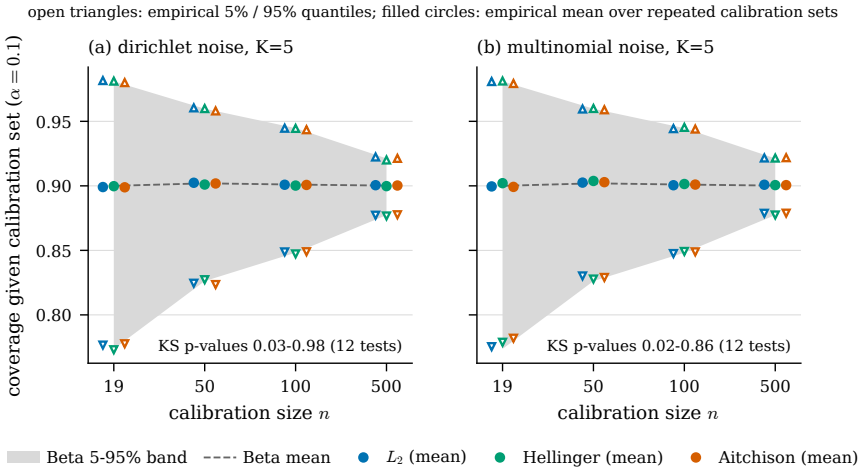


Figure 1: Calibration-conditional coverage of split conformal prediction ($\alpha = 0.1$, $K = 5$, linear predictor) over 3000 calibration sets per cell, for L_2, Hellinger and Aitchison scores. Shaded: exact 5–95% band of $\mathrm{Beta}(k, n+1-k)$; markers: empirical mean and 5/95% quantiles. KS p-values over the 12 cells of each panel are annotated.

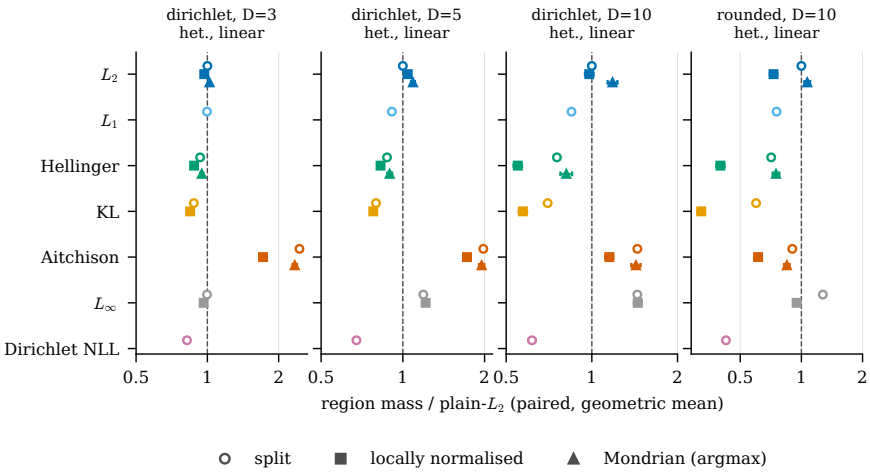


Figure 2: Region mass relative to the plain L_2 region of the same replicate (geometric mean over 100 replicates; 95% t-interval of the log ratio, mostly smaller than the markers), $\alpha = 0.1$, $n_{\mathrm{cal}} = 500$, heteroscedastic, linear predictor. Below 1: more informative at equal coverage. Open: plain split; squares: locally normalised; triangles: Mondrian (argmax).

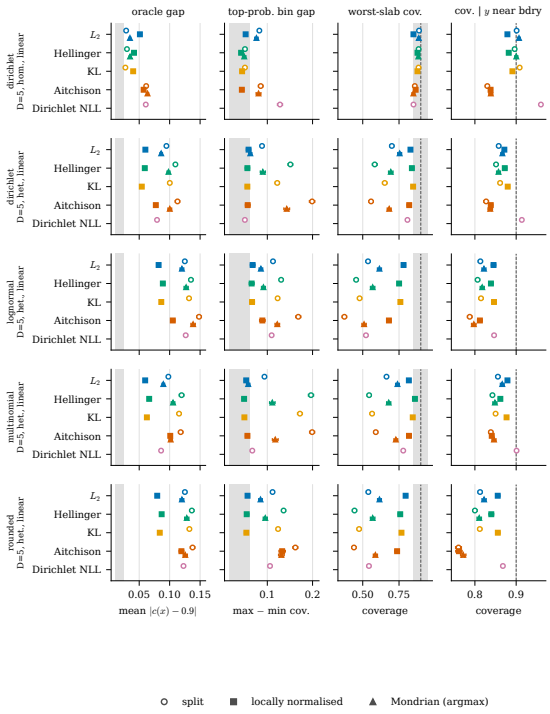


Figure 3: Conditional-coverage metrics in five $K = 5$ worlds ($n_{\text{cal}} = 500$, linear predictor, 100 replicates; the 95% intervals are smaller than the markers). Grey: 5–95% band of the metric when conditional coverage is exact (Fig. 4); dashed: nominal 0.9. Markers as in Fig. 2.

scores follow (KL 0.88, 0.80, 0.70; Hellinger 0.93, 0.87, 0.75), whereas L_∞ is no better than L_2 and worse for larger K (1.19, 1.44 at $K = 5, 10$). (iii) Local normalisation can shrink regions strongly at $K = 10$ (normalised KL: 0.57 of plain L_2 for Dirichlet noise and 0.32 for the rounded world). Table 7 (Appendix C) gives all variants for $K = 5$.

4.4 Conditional coverage: normalisation helps under heteroscedasticity only (C3)

Fig. 3 shows four conditional-coverage metrics at $K = 5$, $n_{\text{cal}} = 500$, with the null band of Fig. 4 in grey: the oracle gap $\overline{|c(x) - 0.9|}$ from the known generator, the gap between the best and worst of five bins of the predicted top probability, worst-slab coverage, and coverage given y near the boundary of the simplex (outcome-conditional and descriptive only). Under homoscedastic Dirichlet noise the oracle gap of the L_2, L_1, L_∞, Hellinger and KL scores is 0.026–0.030, just above the null band (0.011–0.025), and local normalisation does not help (over the 24 homoscedastic configurations the normalised score has the smaller gap in only 8, with means 0.070 against 0.062). Under heteroscedastic noise every plain score is far from the null: the oracle gap is 0.079–0.148 and worst-slab coverage as low as 0.37–0.60, e.g. plain L_2 0.118 and 0.60 against 0.083 and 0.82 for its normalised variant, which is better in all 24 heteroscedastic configurations. Even the best variant (Dirichlet: normalised KL, gap 0.054, wsc 0.85; logistic-normal: normalised L_∞, 0.077, 0.81) stays far above the floor, so normalisation by an estimated scale reduces but does not remove conditional miscoverage. The plain Aitchison score has the lowest worst-slab coverage in three of the four heteroscedastic rows (0.56 Dirichlet, 0.37 logistic-normal, 0.44 rounded; in the multinomial row Hellinger 0.54 and Aitchison 0.59 are lowest), and the plain Dirichlet NLL score is the best plain score only in its own world (0.079, wsc 0.81; in the logistic-normal world its gap equals that of L_2, 0.126 against 0.125). Mondrian regions by argmax of the prediction improve plain scores less than scale normalisation (Table 7). So no score is uniformly the best on both size and conditional coverage: the smallest mass is achieved by the Dirichlet NLL in its own world (0.11) but by normalised Aitchison (0.21, with a gap 0.105) in the logistic-normal world.

4.5 Calibration size and calibration scheme (C4)

Fig. 7 (Appendix C) varies $n_{\mathrm{cal}} \in \{30, 100, 250, 500, 2000\}$ ($K = 5$, heteroscedastic, linear predictor). The standard deviation of test coverage over replicates follows $\sqrt{\mathrm{Var}\,\mathrm{Beta}(k, n+1-k) + 0.09/2000}$, the Beta spread plus test-set noise, for all three scores shown, so the finite-sample coverage fluctuation is score-free also in this setting and shrinks from about 0.05 to 0.01. Region size behaves differently by scheme. Mondrian calibration by argmax needs enough points per group: at $n_{\mathrm{cal}} = 30$ the threshold is infinite for 47% of test points (the region is the whole simplex), coverage is 0.96–0.95, and the median region volume is 0.73 (Dirichlet) and 0.90 (logistic-normal) against 0.13 and 0.53 for plain split; the difference is gone at $n_{\mathrm{cal}} = 250$ in the Dirichlet world, and at $n_{\mathrm{cal}} = 2000$ Mondrian is slightly smaller than plain split (0.09 against 0.11; 0.38 against 0.42) but, for logistic-normal noise, not than the normalised variant. The locally normalised variant is stable in n_{cal} in these worlds; in the logistic-normal world it halves the region (0.21 against 0.53 at $n_{\mathrm{cal}} = 30$, 0.18 against 0.42 at 2000). CV+ (L_2) has near-constant volume (0.10 and 0.42) with coverage 0.90 at every n_{cal}; because it pools fitting and calibration data, its n_{cal} is not on the same footing as split conformal prediction, and CV+ was run for the linear predictor only.

4.6 Zero replacement for the Aitchison score

The Aitchison score needs parts replaced by $\varepsilon > 0$. Marginal coverage is unaffected (0.896–0.909 for all plotted variants) but the region mass of the plain score falls from 0.52–0.63 at $\varepsilon = 10^{-4}$ to 0.13–0.31 at $\varepsilon = 0.1$ (Dirichlet $0.52 \to 0.14$, multinomial $0.57 \to 0.13$, rounded $0.63 \to 0.31$), while worst-slab coverage moves little and non-monotonically and stays below that of normalised L_2 (Fig. 6, Appendix). The zero-replacement constant is therefore a tuning parameter of efficiency that must be reported; here the main grid used $\varepsilon = 10^{-3}$.

4.7 Label-set bridge on CIFAR-10 (C7)

Table 1 gives results at $T = 1$. All methods reach 0.900 marginal coverage, as Proposition 1 implies. As Proposition 2 states, L_1, Hellinger and KL sets are identical to LAC sets in 100% of test points (column "=LAC"), so these three are not different methods on classifier outputs; the Brier (L_2) and Aitchison sets differ from LAC in 2.8% and 2–9% of points at 12 epochs. LAC has the smallest sets (1.055) but the largest class-conditional undercoverage (minimum per-class coverage 0.800 against 0.872 for APS at size 1.303), which is the usual efficiency–adaptivity trade-off [Angelopoulos et al., 2021]. Which score is better on SSCV depends on the checkpoint and zero replacement: with $\varepsilon = 10^{-3}$ the Aitchison score reaches SSCV† 0.911 at 12 epochs but produces 2.6% empty sets and a lower minimum class coverage (0.794), whereas $\varepsilon = 0.1$ gives 0.814; the Aitchison score on a softmax is thus sensitive to the zero-replacement constant.

4.8 Conditional coverage under Dirichlet noise: where the first-order theory holds (C6)

Fig. 8 (Appendix C) compares the delta-method prediction of Section 3.3 with simulated pointwise coverage (oracle mean predictor, 300 values of x per panel, precision $\phi \in \{200, 50, 20, 8\}$). For the Euclidean score the prediction is accurate (RMSE 0.008–0.010 across all four precisions; correlation ≥ 0.94). For the Hellinger score the first-order prediction is exactly flat at 0.9 (the Gaussian approximation is variance-stabilised, which is why no correlation is defined), while the simulated coverage varies in $[0.81, 0.99]$ with RMSE 0.014–0.038 growing as ϕ falls; we only learn that Hellinger conditional coverage is much closer to homogeneous than Euclidean, not by how much. For the Aitchison score the approximation fails (RMSE 0.16–0.27; correlation 0.58 at $\phi = 200$ falling to -0.44 at $\phi = 8$, also for a second-order moment variant): we conjecture that the log-ratio score is dominated by the smallest parts, where the Gaussian approximation of a Dirichlet is poor, but we did not test this. We therefore claim the first-order prediction only for the Euclidean score in the tested range $\phi \geq 8$, $K = 5$.

4.9 How large must a difference in conditional-coverage metrics be? (C5)

Binned and worst-slab metrics are noisy and, being extrema over bins or slabs, biased even for a method with exact conditional coverage. We quantify this by a null simulation: for each n_{cal} we

Table 1: CIFAR-10 label sets, $T = 1$, $n_{\text{cal}} = 1000$, 200 random calibration/test splits of the 10,000 test images; mean $\pm$ half-width of the 95% t-interval over splits. SSCV†: minimum coverage over size strata $\{1\}, \{2\text{–}3\}, \{4+\}$ (strata with fewer than 30 points dropped). "=LAC": share of test points whose set equals the LAC set.

method	cov.	size	min class cov.	SSCV†	empty	= LAC
CNN, 12 epochs (acc. 0.881)						
LAC	0.900 ± 0.001	1.055 ± 0.004	0.800 ± 0.003	0.863 ± 0.002	0.005 ± 0.001	–
LAC, class-cond.	0.904 ± 0.001	1.131 ± 0.005	0.851 ± 0.003	0.842 ± 0.001	0.015 ± 0.001	–
APS	0.900 ± 0.001	1.303 ± 0.005	0.872 ± 0.002	0.898 ± 0.002	0.051 ± 0.001	0.754
RAPS	0.900 ± 0.001	1.285 ± 0.004	0.871 ± 0.002	0.882 ± 0.002	0.050 ± 0.001	0.754
Brier / L_2	0.900 ± 0.001	1.060 ± 0.004	0.801 ± 0.002	0.803 ± 0.003	0.001 ± 0.000	0.972
L_1	0.900 ± 0.001	1.055 ± 0.004	0.800 ± 0.003	0.863 ± 0.002	0.005 ± 0.001	1.000
Hellinger	0.900 ± 0.001	1.055 ± 0.004	0.800 ± 0.003	0.863 ± 0.002	0.005 ± 0.001	1.000
KL	0.900 ± 0.001	1.055 ± 0.004	0.800 ± 0.003	0.863 ± 0.002	0.005 ± 0.001	1.000
Aitchison (ε=0.1)	0.900 ± 0.001	1.059 ± 0.004	0.800 ± 0.002	0.814 ± 0.002	0.001 ± 0.000	0.980
Aitchison (ε=10^{-3})	0.900 ± 0.001	1.074 ± 0.004	0.794 ± 0.003	0.911 ± 0.001	0.026 ± 0.001	0.914
CNN, 2 epochs (acc. 0.740)						
LAC	0.900 ± 0.001	1.762 ± 0.011	0.834 ± 0.002	0.873 ± 0.002	0.000 ± 0.000	–
LAC, class-cond.	0.905 ± 0.001	1.858 ± 0.012	0.852 ± 0.003	0.881 ± 0.002	0.000 ± 0.000	–
APS	0.900 ± 0.001	2.136 ± 0.009	0.877 ± 0.001	0.903 ± 0.001	0.026 ± 0.000	0.609
RAPS	0.900 ± 0.001	2.096 ± 0.009	0.875 ± 0.001	0.885 ± 0.001	0.023 ± 0.000	0.619
Brier / L_2	0.900 ± 0.001	1.844 ± 0.016	0.845 ± 0.002	0.883 ± 0.002	0.000 ± 0.000	0.766
L_1	0.900 ± 0.001	1.762 ± 0.011	0.834 ± 0.002	0.873 ± 0.002	0.000 ± 0.000	1.000
Hellinger	0.900 ± 0.001	1.762 ± 0.011	0.834 ± 0.002	0.873 ± 0.002	0.000 ± 0.000	1.000
KL	0.900 ± 0.001	1.762 ± 0.011	0.834 ± 0.002	0.873 ± 0.002	0.000 ± 0.000	1.000
Aitchison (ε=0.1)	0.901 ± 0.001	1.774 ± 0.011	0.836 ± 0.002	0.872 ± 0.002	0.000 ± 0.000	0.904
Aitchison (ε=10^{-3})	0.900 ± 0.001	1.906 ± 0.011	0.797 ± 0.003	0.891 ± 0.001	0.005 ± 0.000	0.685

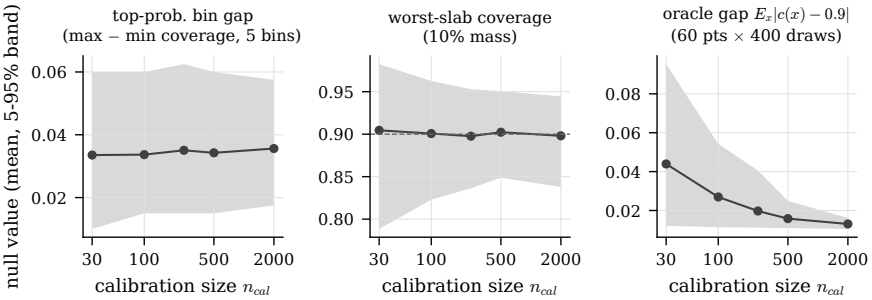


Figure 4: Null distribution of three conditional-coverage metrics when conditional coverage is exact by construction ($n_{\text{test}} = 2000$, 500 replicates per n_{cal}). Line: mean; band: 5–95%.

draw 500 replicates in which the coverage level is $p \sim \text{Beta}(k, n + 1 - k)$ (the exact split-conformal law) and coverage indicators on 2000 test points are Bernoulli(p) independent of the features, then apply the same binning and slab code as in E1 (Fig. 4). At $n_{\text{cal}} = 500$ the null gap between the highest and lowest of 5 equal-mass bins of the predicted maximal part is 0.034 (sd 0.013), the null worst-slab coverage is 0.90 (sd 0.03), and the oracle pointwise gap (60 points, 400 draws each) has a floor of 0.016 (sd 0.005) from Monte-Carlo noise and the calibration-induced deviation of p from 0.9. A difference between methods below these spreads is not interpretable, and we report conditional-coverage gaps in the following section only relative to this null band.

4.10 A real composition-valued response: CIFAR-10H (C8)

CIFAR-10H gives, for each CIFAR-10 test image, the distribution of human labels, a composition with 80.6% exact zeros; the predictor is the 12-epoch CNN softmax ($T = 1$). Because of the zeros and discrete vote proportions, scores tie and Proposition 1 guarantees only $\geq 1 - \alpha$; coverage refers to the noisy vote proportions, not to a latent label law. Dimension is $K = 10$, so the Lebesgue volume is useless; as efficiency we use "mass", the share of the other images' human distributions that fall inside the region (lower is more informative at equal coverage). Table 6 (Appendix C; 100

random splits into 2000/1000/rest for $\hat{\sigma}$, calibration and test) shows: (i) all 18 variants reach marginal coverage 0.900–0.906, slightly above 0.9 for Mondrian (0.905–0.906), consistent with ties and small groups; (ii) plain and Mondrian regions have similar mass (0.103–0.118) for all scores, whereas locally normalised regions are 3–5 times less informative (0.32–0.56) and do not repair conditional coverage; (iii) coverage in the worst of five confidence bins of the CNN stays at 0.63–0.75 for every variant, far below 0.9 and far outside the null band of Fig. 4 (gap 0.034); the entropy of the human distribution is an outcome-conditional diagnostic, not a conditioning variable of the guarantee, so its bin gaps need not vanish. We cannot rank scores on this dataset beyond this: the intervals of several rows overlap, and the single predictor, single dataset and tie structure limit what transfers.

4.11 Covariate shift: a bounded check (C10)

Section 5 excludes shift from the guarantees, but we check one family: $K = 5$, heteroscedastic, linear predictor, calibration $x \sim \mathcal{N}(0, I)$ and test $x \sim \mathcal{N}(\delta w, I)$ with w the direction of increasing noise, $n_{\mathrm{cal}} = 500$, 100 replicates. Fig. 9 (Appendix C) shows that ordinary split conformal prediction loses coverage as δ grows: at $\delta = 1.5$ the plain scores fall to 0.49–0.73 (logistic-normal 0.49–0.59) and the locally normalised ones to 0.70–0.92, so normalisation mitigates but does not remove the loss. Weighted split conformal prediction with the exact likelihood ratio [Tibshirani et al., 2019] restores coverage to 0.895–0.920 in all families and shifts, at the price of a smaller effective calibration size (down to 73 at $\delta = 1.5$). The exact likelihood ratio is known here; with estimated weights this is not guaranteed. This is one shift family on one predictor and supports no general statement about shift.

5 Discussion and limitations

What is score-free and what is not. The finite-sample marginal coverage law is score-free (Proposition 1, Fig. 1); it is the right first test of any evaluation pipeline, and it fails to discriminate between scores. Everything a user may care about beyond it, namely region size, coverage across the simplex and robustness to shift, depends on the score geometry, the noise law and the calibration scheme, and on classifier outputs several of the simplex scores collapse to the same sets (Proposition 2, Table 1). Conditional-coverage metrics computed from bins or slabs have a null spread (Fig. 4) that must be reported next to any comparison.

Scope of the conclusions. Results hold for exchangeable data; for the tested dimensions $K \in \{3, 5, 10\}$ and calibration sizes; for the generators described in Section 4 (fixed regression function per configuration, Gaussian covariates, two predictors); and for one CNN on CIFAR-10 and its CIFAR-10H annotations. They should not be extrapolated to (i) $K > 10$ or to Lebesgue-volume comparisons for $K \geq 10$, where we use a different, dimension-valid measure (mass of a reference law) and compare scores only within it; (ii) full conformal or jackknife+ with neural predictors, which we did not run (CV+ is run only for the Euclidean score with the linear predictor); (iii) covariate or label shift beyond the single family in Section 4; (iv) batch multivalid or other group-conditional methods [Jung et al., 2022], which we did not run; (v) response distributions with heavy mass on the boundary other than those generated, in particular the zero-heavy CIFAR-10H proportions, where ties give only a lower bound.

Limitations. We claim no new coverage guarantee and no conditional guarantee: exact conditional coverage is unattainable without assumptions, and the oracle measurements are specific to the generators. The delta-method prediction is first-order and fails for the Aitchison score at the tested precisions (Fig. 8). The Aitchison and Dirichlet-likelihood scores depend on the zero-replacement constant ε, which we varied only in a sensitivity run. Efficiency depends on the reference measure (Lebesgue on the simplex, or the marginal law of y); rankings under the two may differ and we do not compare across them. RAPS is not tuned. Intervals reflect variability over data and calibration draws for fixed regression functions; they do not account for variation over the choice of regression function, since each configuration fixes one with seed 0. The synthetic worlds were designed by us; their noise levels (e.g. $\phi_0 = 20$) determine how sharp the differences between scores are. Our novelty claim is limited to the literature we read (Section 2); we did not search exhaustively.

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A Reproduction notes

All code, result files and run commands are in the accompanying repository: `src/` (library), `scripts/` (experiment runners), `results/` (CSV files and a column guide), `figtab/` (one script per figure and table, with input files recorded in `figtab/MANIFEST.md`) and `REPRODUCE.md` (environment and package versions, data sources and versions, seeds, commands, expected run times, and the map from each exhibit to its script). Data: CIFAR-10 (copy from the Hugging Face dataset `uoft-cs/cifar10`, a fixed revision recorded in `REPRODUCE.md`; no licence stated by the source) and CIFAR-10H human label distributions (no licence stated); both are cited in `REPRODUCE.md`. No forbidden material was encountered; this is recorded there.

Exhibit map. Numbering in the paper differs from the file names in `figures/` and `tables/`: Fig. 1 `fig1_beta_law`; Fig. 2 `fig2_size`; Fig. 3 `fig3_conditional`; Fig. 4 `fig5_null_spread`; Fig. 7 `fig4_ncal`; Fig. 8 `fig6_delta_method`; Fig. 9 `fig7_shift`; Fig. 5 `fig2_size_volume`; Fig. 6 `fig8_eps_sensitivity`; Table 7 `tab2_e1_headline`; Table 1 `tab3_e2_labelsets`; Table 6 `tab4_e3_cifar10h`; Tables 2–5 `tabA_e1_grid_*`. Each is produced by the script of the same stem in `figtab/` from the result files named in `figtab/MANIFEST.md`.

B Additional E1 exhibits

C Further exhibits for Section 4

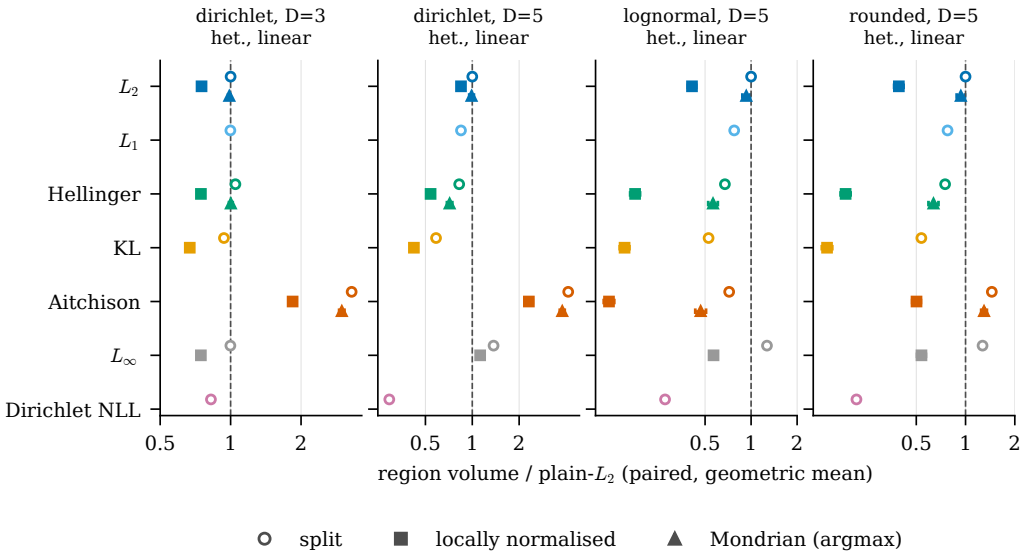


Figure 5: As Fig. 2 with the Lebesgue volume of the region (Monte-Carlo) instead of the mass, in four configurations with $K \leq 5$ (Dirichlet $K = 3, 5$; logistic-normal and rounded $K = 5$), where volume is computable.

Table 2: All 12 Dirichlet-noise configurations of the E1 grid ($n_{\text{cal}} = 500$, 100 replicates; means; largest 95% half-widths: oracle gap 0.004, wsc 0.015, mass 0.01). cov. range: minimum–maximum over the 16 methods of mean coverage. Oracle gap and wsc for plain and normalised L_2; mass for plain L_2, KL, Aitchison and Dirichlet NLL. Logistic-normal, multinomial and rounded: Tables 3–5.

				oracle gap		worst slab		region mass			
D	het.	pred.	cov. range	L_2	L_2-n	L_2	L_2-n	L_2	KL	Ait.	NLL
3	no	linear	0.899–0.903	0.049	0.057	0.87	0.85	0.153	0.126	0.352	0.125
3	no	rf	0.899–0.903	0.092	0.084	0.82	0.84	0.235	0.216	0.586	0.193
3	yes	linear	0.900–0.904	0.093	0.057	0.72	0.86	0.177	0.155	0.434	0.145
3	yes	rf	0.899–0.905	0.111	0.090	0.71	0.83	0.262	0.244	0.630	0.211
5	no	linear	0.897–0.903	0.029	0.051	0.89	0.85	0.137	0.102	0.272	0.107
5	no	rf	0.896–0.904	0.070	0.069	0.82	0.85	0.209	0.168	0.404	0.169
5	yes	linear	0.896–0.903	0.095	0.060	0.70	0.83	0.170	0.135	0.335	0.114
5	yes	rf	0.896–0.902	0.098	0.082	0.73	0.82	0.239	0.197	0.449	0.171
10	no	linear	0.898–0.909	0.035	0.062	0.89	0.82	0.061	0.041	0.095	0.049
10	no	rf	0.898–0.908	0.094	0.091	0.82	0.82	0.126	0.089	0.188	0.103
10	yes	linear	0.898–0.908	0.107	0.067	0.62	0.82	0.102	0.072	0.148	0.063
10	yes	rf	0.897–0.908	0.114	0.102	0.65	0.80	0.171	0.126	0.245	0.117

Table 3: As Table 2, logistic-normal noise.

				oracle gap		worst slab		region mass			
D	het.	pred.	cov. range	L_2	L_2-n	L_2	L_2-n	L_2	KL	Ait.	NLL
3	no	linear	0.901–0.904	0.056	0.063	0.88	0.84	0.228	0.175	0.165	0.171
3	no	rf	0.898–0.904	0.073	0.078	0.87	0.85	0.297	0.255	0.333	0.234
3	yes	linear	0.898–0.903	0.123	0.074	0.57	0.85	0.359	0.326	0.421	0.293
3	yes	rf	0.899–0.904	0.126	0.092	0.56	0.85	0.422	0.399	0.543	0.346
5	no	linear	0.899–0.905	0.032	0.055	0.90	0.85	0.204	0.119	0.078	0.111
5	no	rf	0.898–0.906	0.058	0.064	0.86	0.86	0.274	0.190	0.200	0.166
5	yes	linear	0.901–0.905	0.125	0.082	0.54	0.78	0.431	0.323	0.346	0.250
5	yes	rf	0.900–0.906	0.127	0.093	0.52	0.83	0.486	0.385	0.455	0.280
10	no	linear	0.899–0.908	0.044	0.070	0.88	0.80	0.076	0.027	0.010	0.027
10	no	rf	0.897–0.907	0.085	0.084	0.83	0.82	0.149	0.078	0.091	0.064
10	yes	linear	0.898–0.909	0.138	0.082	0.45	0.79	0.265	0.158	0.160	0.117
10	yes	rf	0.898–0.908	0.137	0.107	0.47	0.78	0.331	0.224	0.263	0.144

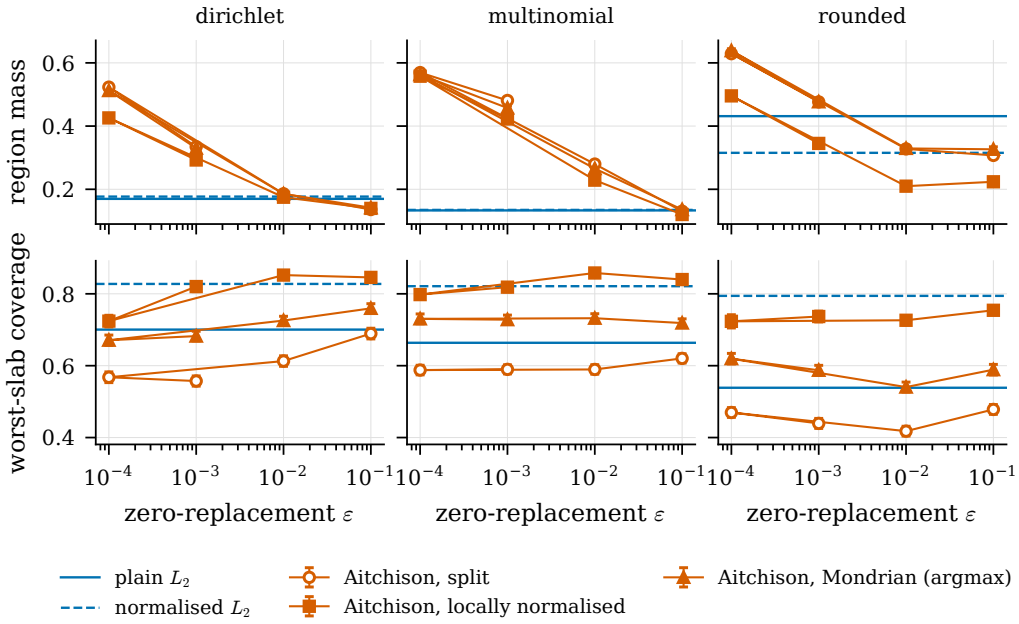


Figure 6: Sensitivity of the plain Aitchison score to the zero replacement ε ($K = 5$, heteroscedastic, linear predictor, $n_{\text{cal}} = 500$, 100 replicates, 95% t-intervals). Top: region mass; bottom: worst-slab coverage. Blue lines: plain and normalised L_2, independent of ε.

Table 4: As Table 2, multinomial noise.

				oracle gap		worst slab		region mass			
D	het.	pred.	cov. range	L_2	L_2-n	L_2	L_2-n	L_2	KL	Ait.	NLL
3	no	linear	0.900–0.904	0.051	0.057	0.88	0.87	0.117	0.103	0.607	0.103
3	no	rf	0.900–0.904	0.107	0.093	0.80	0.81	0.209	0.200	0.751	0.172
3	yes	linear	0.900–0.904	0.099	0.061	0.70	0.88	0.140	0.138	0.648	0.129
3	yes	rf	0.900–0.905	0.119	0.100	0.70	0.82	0.226	0.227	0.739	0.197
5	no	linear	0.898–0.907	0.029	0.049	0.90	0.86	0.097	0.079	0.416	0.103
5	no	rf	0.897–0.907	0.086	0.079	0.79	0.83	0.173	0.148	0.503	0.172
5	yes	linear	0.900–0.907	0.098	0.060	0.66	0.82	0.133	0.132	0.481	0.127
5	yes	rf	0.900–0.905	0.103	0.090	0.72	0.81	0.207	0.195	0.549	0.188
10	no	linear	0.898–0.907	0.036	0.061	0.89	0.83	0.038	0.028	0.159	0.060
10	no	rf	0.897–0.906	0.115	0.104	0.80	0.80	0.102	0.073	0.234	0.126
10	yes	linear	0.899–0.908	0.113	0.063	0.57	0.82	0.071	0.066	0.253	0.085
10	yes	rf	0.898–0.909	0.123	0.114	0.64	0.77	0.141	0.119	0.335	0.153

Table 5: As Table 2, rounded logistic-normal noise.

				oracle gap		worst slab		region mass			
D	het.	pred.	cov. range	L_2	L_2-n	L_2	L_2-n	L_2	KL	Ait.	NLL
3	no	linear	0.900–0.903	0.055	0.061	0.88	0.85	0.227	0.174	0.228	0.170
3	no	rf	0.899–0.905	0.073	0.078	0.87	0.85	0.296	0.256	0.414	0.232
3	yes	linear	0.898–0.902	0.123	0.074	0.57	0.85	0.358	0.327	0.583	0.290
3	yes	rf	0.899–0.903	0.126	0.092	0.56	0.85	0.422	0.401	0.743	0.344
5	no	linear	0.899–0.905	0.032	0.053	0.89	0.85	0.203	0.118	0.279	0.110
5	no	rf	0.898–0.906	0.058	0.062	0.86	0.87	0.274	0.189	0.428	0.164
5	yes	linear	0.900–0.906	0.125	0.080	0.54	0.80	0.431	0.325	0.476	0.239
5	yes	rf	0.900–0.907	0.127	0.093	0.52	0.83	0.485	0.386	0.580	0.274
10	no	linear	0.898–0.907	0.045	0.067	0.88	0.82	0.075	0.026	0.078	0.028
10	no	rf	0.897–0.906	0.085	0.082	0.82	0.82	0.148	0.077	0.193	0.070
10	yes	linear	0.898–0.910	0.138	0.079	0.45	0.80	0.262	0.157	0.236	0.112
10	yes	rf	0.898–0.907	0.137	0.105	0.47	0.80	0.330	0.223	0.324	0.152

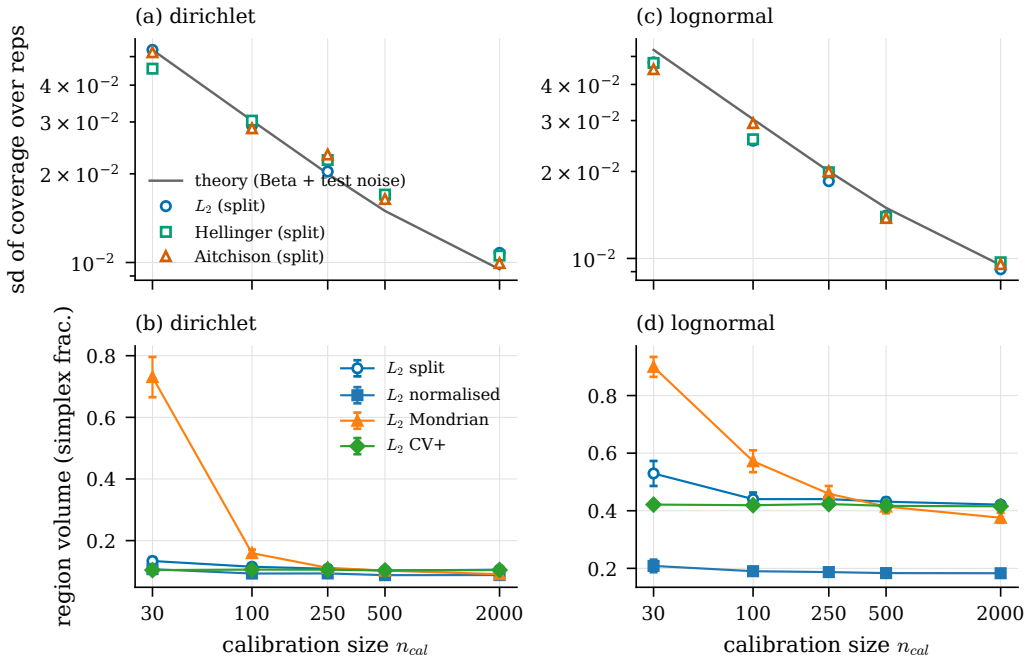


Figure 7: Calibration size ($K = 5$, heteroscedastic, linear predictor, 100 replicates per point). (a,c) Standard deviation over replicates of test coverage; line: $\sqrt{\mathrm{Var}\,\mathrm{Beta} + 0.09/2000}$. (b,d) Median region volume (fraction of the simplex) with 95% t-intervals for split, locally normalised and Mondrian L_2 and for CV+; $n_{\mathrm{cal}} = 500$ points come from the main grid.

Table 6: CIFAR-10H as a composition-valued response, 12-epoch CNN softmax, $\alpha = 0.1$, $n_{\mathrm{cal}} = 1000$, 100 random splits; mean $\pm$ half-width of the 95% t-interval. mass: share of 300 other images' human distributions inside the region; ambig. min / conf. min: coverage in the worst of 5 equal-mass bins of the entropy of the human distribution / of the CNN's top-1 probability; wsc: worst-slab coverage ($\delta = 0.1$).

score	scheme	cov.	mass	ambig. min	conf. min	wsc
L_2	split	0.901 ± 0.002	0.106 ± 0.001	0.810 ± 0.005	0.660 ± 0.008	0.658 ± 0.009
	normalised	0.901 ± 0.003	0.394 ± 0.010	0.785 ± 0.005	0.684 ± 0.004	0.671 ± 0.007
	Mondrian	0.906 ± 0.002	0.116 ± 0.001	0.823 ± 0.004	0.694 ± 0.006	0.691 ± 0.009
L_1	split	0.901 ± 0.002	0.106 ± 0.001	0.801 ± 0.005	0.659 ± 0.008	0.624 ± 0.008
	normalised	0.901 ± 0.002	0.399 ± 0.010	0.787 ± 0.005	0.685 ± 0.004	0.672 ± 0.007
	Mondrian	0.906 ± 0.002	0.112 ± 0.001	0.816 ± 0.004	0.689 ± 0.007	0.664 ± 0.007
L_∞	split	0.901 ± 0.002	0.106 ± 0.001	0.811 ± 0.004	0.660 ± 0.008	0.636 ± 0.009
	normalised	0.901 ± 0.002	0.396 ± 0.010	0.788 ± 0.005	0.686 ± 0.004	0.677 ± 0.006
	Mondrian	0.905 ± 0.002	0.115 ± 0.001	0.824 ± 0.004	0.689 ± 0.007	0.669 ± 0.009
Hellinger	split	0.901 ± 0.002	0.104 ± 0.001	0.821 ± 0.004	0.652 ± 0.009	0.611 ± 0.010
	normalised	0.901 ± 0.003	0.322 ± 0.009	0.789 ± 0.005	0.690 ± 0.005	0.680 ± 0.007
	Mondrian	0.906 ± 0.002	0.110 ± 0.001	0.832 ± 0.004	0.682 ± 0.007	0.657 ± 0.008
KL	split	0.901 ± 0.002	0.103 ± 0.001	0.758 ± 0.005	0.711 ± 0.008	0.669 ± 0.009
	normalised	0.901 ± 0.002	0.319 ± 0.009	0.772 ± 0.005	0.673 ± 0.004	0.666 ± 0.006
	Mondrian	0.905 ± 0.002	0.109 ± 0.001	0.769 ± 0.005	0.745 ± 0.006	0.717 ± 0.008
Aitchison	split	0.901 ± 0.002	0.110 ± 0.001	0.817 ± 0.004	0.626 ± 0.008	0.570 ± 0.010
	normalised	0.900 ± 0.002	0.559 ± 0.014	0.739 ± 0.005	0.649 ± 0.003	0.642 ± 0.007
	Mondrian	0.905 ± 0.002	0.118 ± 0.001	0.831 ± 0.003	0.668 ± 0.007	0.630 ± 0.008

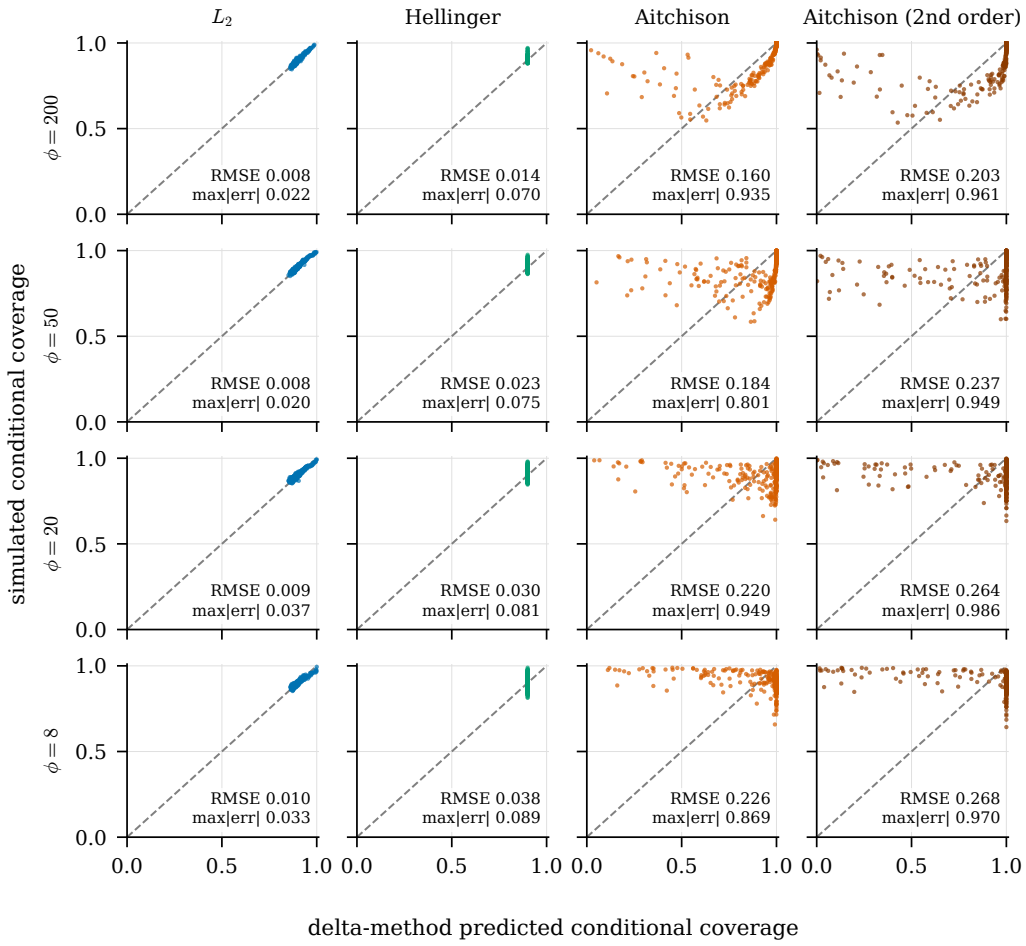


Figure 8: Delta-method prediction (horizontal) against simulated pointwise conditional coverage (vertical) of the marginal threshold at nominal 0.9, $K = 5$, Dirichlet noise, oracle mean predictor. Rows: precision ϕ; columns: score (Aitchison in first-order and second-order moment versions). Dashed: identity.

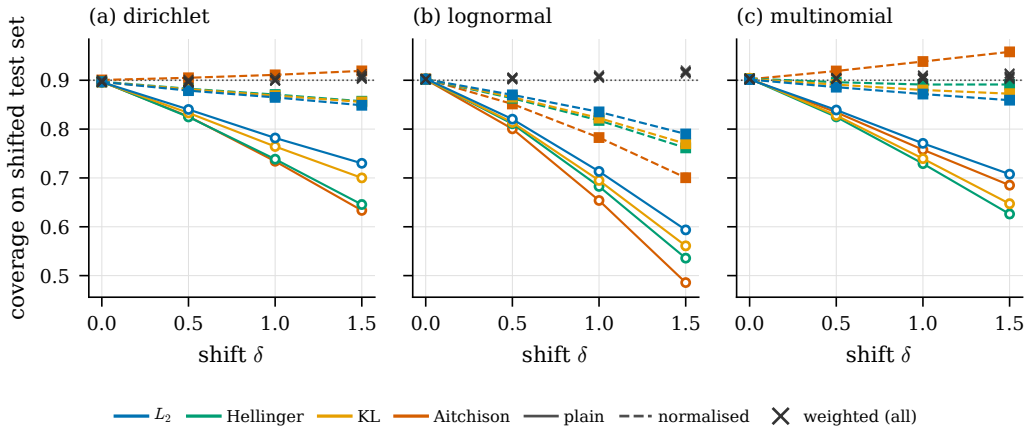


Figure 9: Test coverage under covariate shift of size δ towards noisier inputs ($K = 5$, heteroscedastic, linear predictor, $n_{\text{cal}} = 500$, 100 replicates). Lines: ordinary split conformal (solid: plain, dashed: locally normalised) with 95% t-intervals; crosses: weighted split conformal with the exact likelihood ratio.

Table 7: E1, $K = 5$, heteroscedastic, linear predictor, $n_{\text{cal}} = 500$, means over 100 replicates (largest 95% half-widths: coverage 0.004, wsc 0.016, oracle gap 0.004, mass 0.009). wsc: worst-slab coverage; or. gap: oracle conditional-coverage gap (null floor 0.011–0.025); mass: region mass. Bold: best of the 15 methods in the column (no significance claim).

	dirichlet, D=5, het., linear				lognormal, D=5, het., linear			
score	cov.	wsc	or. gap	mass	cov.	wsc	or. gap	mass
plain								
L_2	0.896	0.70	0.095	0.170	0.903	0.54	0.125	0.431
L_1	0.898	0.71	0.092	0.154	0.902	0.52	0.126	0.374
L_∞	0.896	0.70	0.094	0.202	0.902	0.54	0.124	0.490
Hellinger	0.897	0.58	0.109	0.148	0.902	0.45	0.135	0.354
KL	0.896	0.65	0.100	0.135	0.902	0.48	0.132	0.323
Aitchison	0.897	0.56	0.113	0.335	0.902	0.37	0.148	0.346
Dirichlet NLL	0.898	0.81	0.079	**0.114**	0.903	0.52	0.126	0.250
locally normalised								
L_2	0.897	0.83	0.060	0.177	0.902	0.78	0.082	0.318
L_∞	0.897	0.84	0.057	0.206	0.902	**0.81**	**0.077**	0.358
Hellinger	0.897	0.84	0.059	0.141	0.902	0.75	0.089	0.244
KL	0.896	**0.85**	**0.054**	0.132	0.902	0.76	0.086	0.228
Aitchison	0.901	0.82	0.077	0.293	0.903	0.68	0.105	**0.209**
Mondrian								
L_2	0.901	0.75	0.086	0.185	0.905	0.62	0.120	0.434
Hellinger	0.900	0.69	0.098	0.152	0.905	0.57	0.127	0.360
Aitchison	0.903	0.68	0.100	0.330	0.905	0.51	0.139	0.352

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